

AI LEGAL™

AI Notes Maker & Study Workspace Module

Comprehensive Product, Technical & Architecture Documentation

Document Area	Details
Module Route	/dashboard/tools/notes-maker
Primary Experience	Prompt / Voice Input → Configuration → AI Notes Canvas → Saved Notes Repository
Core Concept	AI-Powered Legal Synthesis, Study Notes & Case Brief Workspace
Target Users	BA LLB / LLB Students, LLM Students, PCS-J, CLAT, AIBE & Nepal Legal Exam Aspirants

Document Structure

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1. Executive Summary & Module Purpose

The AI Notes Maker (/dashboard/tools/notes-maker) is a specialized legal synthesis workspace designed to automate the structuring, summarization, and compilation of complex legal doctrines, judicial precedents, and statutory codes into exam-ready study notes and case briefs.

The module supports different academic and competitive-examination needs by allowing users to select a note structure and academic level before generating content. The resulting output is rendered as structured legal Markdown that can be edited, saved, copied, searched, and exported.

The workspace is designed to bridge the gap between raw legal material and practical revision material by combining AI synthesis, structured formatting, voice input, personal annotations, and a persistent study vault.

2. Core Value Propositions

Value Proposition	Description
6 Specialized Pedagogical Formats	Generates Short Notes, Detailed Notes, Revision Notes, Exam Notes, Case Summaries, and Bare Act Summaries according to the user's study objective.
Academic Level Calibration	Adjusts analytical depth, terminology, structure, and theoretical rigor for BA LLB, LLB, LLM, Judiciary, CLAT, AIBE, Nepal Bar Council, and Judicial Service preparation.
Multi-Modal Voice & Text Input	Supports typed prompts as well as browser-based Speech-to-Text for hands-free study and lecture/revision workflows.
Structured Legal Markdown Engine	Renders headings, statutory sections, legal ingredients, precedent blocks, procedural guidance, bullets, and other semantic structures.
Personal Study Vault & Export Studio	Allows users to save, tag, pin, organize, search, copy, edit, and export generated notes for continued study.

3. Six Specialized Note Formats

#	Format	Pedagogical Focus & Output Structure	Ideal Use Case
1	Short Notes	1-page synopsis covering definitions, statutory thresholds, core sections, and essential legal principles.	Quick concept grasp and classroom introduction
2	Detailed Notes	In-depth academic treatment covering legislative history, ingredients, exceptions, jurisprudential debates, and comparative law.	Semester exams and thesis/research preparation
3	Revision Notes	High-yield bullets, mnemonic aids, cross-statute comparisons, and quick-reference structures.	Last-week revision and rapid recall
4	Exam Notes	Model-answer structure using Issue, Rule, Application, Conclusion (IRAC), with authority references and answer-oriented organization.	PCS-J Mains and university examinations
5	Case Summary	Law brief containing parties, court/bench, facts, issues, arguments, ratio decidendi, and obiter dicta.	Moot court research and case compilation
6	Bare Act Summary	Clause-by-clause plain-language explanation identifying provisos, conditions, exceptions, and penalties.	Decoding statutes such as BNS / BNSS

4. Architectural & User Workflow Lifecycle

Stage	Workflow
1. Input	Prompt & Topic Box OR Voice / Mic Input
2. Configuration	Select Note Format, Academic Level, and Jurisdiction
3. Generation	Generate AI Notes using configured prompt instructions
4. Interactive Canvas	Review, edit, annotate, copy, save, and export generated notes
5. Saved Repository	Search, pin, organize, revisit, share, or delete saved notes

4.1 Step 1 — Input & Configuration

- Students can enter legal queries such as “Doctrine of Basic Structure”, “Section 103 BNS Murder Ingredients”, or “Bail under Section 439 CrPC”.
- Voice input supports continuous spoken prompts through browser SpeechRecognition.
- Quick-selection topic chips can provide one-click starting points for topics such as Fundamental Rights, Bharatiya Nyaya Sanhita, Cheque Bounce, and Evidence Admissibility.
- Users select the desired note format, academic level, and jurisdiction before generation.

4.2 Step 2 — Generation Engine & Multi-Layer Parsing

- The generation engine invokes the Gemini API using prompt instructions customized to the selected Note Format and Academic Level.
- System instructions can enforce legal accuracy, statutory citation accuracy, and format-specific structural requirements.
- The response passes through sanitization via extractCleanText to remove JSON wrappers, escape characters, or extraneous model tokens.
- Clean semantic Markdown is then rendered in the Notes Canvas.

4.3 Step 3 — Study Workspace Canvas & Tooling

- Copy to Clipboard: Copies the generated note while preserving its Markdown structure.
- Save to Personal Vault: Persists the note locally and supports cloud synchronization where implemented.
- Export Studio: Prepares printer-friendly PDF or document-oriented outputs.
- Append / Edit Mode: Allows students to insert personal classroom commentary, corrections, or additional study material.

5. Technical Implementation Details

Layer	Implementation	Purpose
Web Frontend	NotesMakerWorkspace.jsx	Primary Notes Maker workspace and study canvas.
Mobile Frontend	notes-maker.tsx	Mobile implementation of the Notes Maker experience.
AI Engine	Google Gemini API / generateChatResponse	Generates structured legal notes using format- and level-specific instructions.
Speech-to-Text	SpeechRecognition / webkitSpeechRecognition	Converts spoken prompts into text for note generation.
Voice Locales	en-IN / hi-IN / ne-NP	Supports language-sensitive speech recognition configurations.

Layer	Implementation	Purpose
Saved Notes	student_saved_notes_	Persists the user's saved study notes.
Study Streak	recordStudyActivity	Registers note-generation activity and updates daily study streak.

6. Cross-Module Synergies

- AI Knowledge Hub: Users can read statutory provisions and trigger note synthesis directly without manual copy-pasting.
- Quiz & MCQ Practice: Incorrect answers and explanations can be converted into targeted remediation notes.
- Student Learning Portal: Saved notes can contribute to Tools Completed and Hours Studied metrics.
- Legal Precedents Workspace: Case summaries can serve as a structured entry point for deeper precedent research.
- Study Analytics: Saved notes and generation activity can contribute to personalized study progress.

7. Functional Requirements & UX Behaviour

Requirement Area	Expected Behaviour
Input	Support text and voice-based study queries with clear input states and error handling.
Format Selection	Present six note-format options with clear descriptions of intended use.
Academic Calibration	Allow the user to specify academic or examination level before generation.
Jurisdiction	Allow India/Nepal and other supported jurisdictional contexts to influence the generated content.
Generation	Render clean semantic Markdown and preserve headings, lists, citations, and structured legal content.
Editing	Allow users to append or edit generated notes without losing the original study content.
Persistence	Save notes with user identity, subject/topic metadata, and timestamps where supported.
Export	Provide printer-friendly and document-oriented output options.
Accessibility	Maintain readable typography and usable voice-input controls across desktop and mobile.

8. Persistence & Study Activity Model

The Notes Maker can be represented through a user-centric workflow model:

User → Topic → Input Mode → Academic Level → Note Format → Jurisdiction → AI Generation → Editable Canvas → Saved Note → Study Activity

Saved notes are associated with the user's account through the student_saved_notes_ persistence key. Generation activity is also connected to the Student Streak Service through recordStudyActivity, allowing completed study actions to contribute to the user's daily study streak.

9. Future Extensions

- Version history for notes, enabling users to compare previous edits and restore earlier versions.

- AI-powered note refinement such as “shorten”, “expand”, “convert to exam answer”, and “make revision flashcards”.
- Automatic citation verification against authoritative legal sources.
- Spaced-repetition reminders based on saved notes and revision frequency.
- Collaborative notes and controlled sharing for study groups or classrooms.
- Deeper multilingual support across Indian regional languages and Nepal legal-study contexts.
- Direct conversion of saved notes into Knowledge Hub revision layers and Quiz Practice question sets.

Appendix A — Example User Journey

- 1. User opens AI Notes Maker.
- 2. User enters a legal topic or speaks the query through the microphone.
- 3. User selects note format, academic level, and jurisdiction.
- 4. AI generates a structured legal note using the configured prompt.
- 5. System sanitizes and renders the result as clean Markdown.
- 6. User edits, adds personal commentary, copies, saves, or exports the note.
- 7. Saved content becomes searchable inside the Personal Study Vault.
- 8. Study activity updates the user's learning progress and streak where enabled.

Appendix B — Reference Implementation Paths

Area	Reference
Web	AI-Legal_App_Webapp/src/pages/NotesMakerWorkspace.jsx
Mobile	AI-Legal_App_frontend-mobile/src/app/tools/notes-maker.tsx
AI Engine	generateChatResponse / Google Gemini integration
Persistence	student_saved_notes_
Speech Recognition	SpeechRecognition / webkitSpeechRecognition
Study Activity	recordStudyActivity